v5 construction of R

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1 introduction

The study of the real numbers & its construction in ZFC is well studied and well developed. This paper therefore does not aim to provide an inherently new theorem. Instead, the purpose of this paper is to construct the real numbers under the influence of a certain philisophical guideline. In doing so, my purpose is to demonstrate the beauty in proofs. This is something lean can inherently fundimentally not encode. I wish to demonstrate that there is a part of mathematics that no artificial intelegence or formalization can ever do for you. This is because this part of mathematics is inherent the mathematition him or heself. It is the mathematitions notion of beauty. Beauty is loosely defined, however, the set of all beautiful things seem to have soom loose patterns – beautiful proofs seem simple, or symmetrical. It is precisely this "proof by simular argument" that often makes proofs beautiful, and it is precisely this "proof by simular argument" that does not exist in a formal language. I often consider abstraction beauty, however, this is really just that "simular argument" thing — applying the same argument/definition to a larger class of things (i.e. "abstraction"). The philosophy of constructing the reals: that rational looking sets inherit these kind of psudo metric spaces that are not complete. We wish to "complete" such metric spaces. This completion gives us R. The practise of mathematics will not stop because a machine can now practise it even if this machine is in every way imaginable better at it – the same reason an artist will not stop painting because a diffussion model can make a painting design and then a robotic arm can produce such a painting. Infact, the machine produced paintings can be appreciated by the mathematitions. Mathematics may be as concrete of a science as you can get, yet its practise may require the most art of any other field (creativity, for example). something like this idk.

What makes mathematics mathematics?

Recent publications by open AI of math papers demonstrate the potential of LLMs and AI to discover and write proofs. Can a AI practice mathematics for you? I argue there are parts to mathematics that make it inherent to the practiser – namely the practisers feelings of appreciation and curiosity. On the ladder, a mathemation is in some sense an explorer – let me make up a world in which these are the rules, and then travel through this world and find out what is true and what isn't. What questions do we care about? It is what questions the mathematition cares about. Maybe an AI can explore aswell, but an AI definitionally cannot explore for you. But maybe AI can be your source of fire as you travel into a cold and unknown rhelm. On the former, we often appreciate beauty. What makes a proof or a theorem beautiful? I often find coorilation between beauty and symmetry. The so called "proof of the other case by simular argument" is something lean cannot do, but is something that can make a proof beautiful, because of its symmetry. Maybe it is abstraction (of which seems to be in coorilation with symmetry aswell) that makes mathematics beautiful. Or, maybe, it is a certain philisophical guideline that is followed in a proof or a theorem or list of axioms. In this paper, I try and follow such guideline; namely, that the real numbers are the things you get when you "complete" the rationals, and that the real numbers are themselves sequences of rationals. I don't present anything inherently new, but I try to make what is well studied pretty. I explore what happens when I follow these guidelines, what happens if I use "rational-like" sets instead, and try and abstract my proof to work for completing any metric space. I wish to demonstrate by example that the practise of mathematics carries something very human, at least as human as emotion.

2 Notion of a limit (Review of Topology)

DEFINITION 2.1. (topology) | A *topology* on a set X is a collection τ of subsets of X (the *open sets*) such that

1. $\emptyset \in \tau$ and $X \in \tau$,
2. any union of members of τ is in τ,
3. any finite intersection of members of τ is in τ.

A *neighborhood* of a point x, written $U(x)$, is any open set containing x.

DEFINITION 2.2. (spanning topology) | Say we are given a collection $F \subseteq 2^X$ of subsets of X that we want to be open. I define the *minimal* (or *generated*, or *spanning*) topology of F to be

$$\tau_0(F) := \bigcap \{\, \tau \mid \tau \text{ is a topology on } X \text{ and } F \subseteq \tau \,\}.$$

Intuitively, $\tau_0(F)$ is what results from applying arbitrary unions and finite intersections to the members of F and collecting everything so obtained (hence the name *generated*). Remark 2.4 makes this precise: one round of finite intersections followed by one round of unions suffices.

Lemma 2.3. $\tau_0(F)$ is a topology on X, it contains F, and it is contained in every topology that contains F. So $\tau_0(F)$ is the smallest topology in which every member of F is open.

Remark 2.4. (generated = unions of finite intersections) | Let $F^\cap := \{B_1 \cap \cdots \cap B_k \mid k \geq 0,\ B_i \in F\}$ (the empty intersection, $k = 0$, is X) and $F^{\cap\cup} := \{\text{unions of members of } F^\cap\}$ (the empty union is $\emptyset$). Then $\tau_0(F) = F^{\cap\cup}$:

$$F \subseteq F^\cap \subseteq F^{\cap\cup}; \qquad \emptyset, X \in F^{\cap\cup}; \qquad \textstyle\bigcup_\beta \bigcup_\alpha A_{\beta\alpha} = \bigcup_{\beta,\alpha} A_{\beta\alpha}; \qquad \big(\bigcup_\alpha A_\alpha\big) \cap \big(\bigcup_\beta B_\beta\big) = \bigcup_{\alpha,\beta}(A_\alpha \cap B_\beta)$$

$$\Rightarrow\ F^{\cap\cup} \text{ is a topology containing } F \overset{2.3}{\Longrightarrow} \tau_0(F) \subseteq F^{\cap\cup}.$$

$$\tau \supseteq F \text{ a topology } \Rightarrow\ F^\cap \subseteq \tau\ \Rightarrow\ F^{\cap\cup} \subseteq \tau\ \Rightarrow\ F^{\cap\cup} \subseteq \tau_0(F).$$

The definition of $\tau_0(F)$ is abstract: it does not say which sets are open. When F has two further properties we get a concrete description. This is how we will understand the metric topology: it is the span of the sets we want to be open, namely the open metric balls.

DEFINITION 2.5. (basis) | $F \subseteq 2^X$ is a *basis* if F covers X, and whenever $x \in B_1 \cap B_2$ with $B_1, B_2 \in F$ there is a $B_3 \in F$ with $x \in B_3 \subseteq B_1 \cap B_2$.

THEOREM 2.6. (basis spanning topology) | If F is a basis, then

$$\tau_0(F) \ = \ \{\, U \subseteq X \mid \forall x \in U,\ \exists B \in F \mid x \in B \subseteq U \,\} \ = \ \{\text{unions of members of } F\}.$$

In particular, $U(x)$ is a neighborhood of x in $\tau_0(F)$ exactly when it contains some $B \in F$ with $x \in B$.

THEOREM 2.7. (order topology) | Any totally ordered set $(X, <)$ inherits the following natural topology: the topology spanned by the basis $\mathcal{B}$ consisting of X itself, all open intervals $(a, b) = \{x \mid a < x < b\}$, and all open rays $(a, \rightarrow) = \{x \mid a < x\}$ and $(\leftarrow, b) = \{x \mid x < b\}$.

The space $\mathbb{N} \cup \{\infty\}$. Order $\mathbb{N} \cup \{\infty\}$ by putting ∞ above every natural number, and give it the order topology.

Lemma 2.8. Every neighborhood of ∞ in $\mathbb{N} \cup \{\infty\}$ contains a ray $(N, \rightarrow) = \{n \in \mathbb{N} \mid n > N\} \cup \{\infty\}$ for some $N \in \mathbb{N}$. Every such ray is open, and it contains natural numbers.

Next we define limits in general, and then specialize to sequences. A point $x_0 \in X$ is *adherent* to $E \subseteq X$ if every neighborhood of x_0 meets E. (Every point of E is adherent to E; so is every limit point of E.)

DEFINITION 2.9. (limit) | Let X, Y be topological spaces, $E \subseteq X$, $f : E \to Y$, and let $x_0 \in X$ be adherent to E. Then

$$\lim_{\substack{x \to x_0 \\ x \in E}} f(x) \text{ exists } \iff \exists L \in Y \mid \forall V(L),\ \exists U(x_0) \mid f\big(U(x_0) \cap E\big) \subseteq V(L).$$

Such an L is called a limit. When Y is Hausdorff (Lemma 2.11) the limit is unique and we write $L = \lim f(x)$.

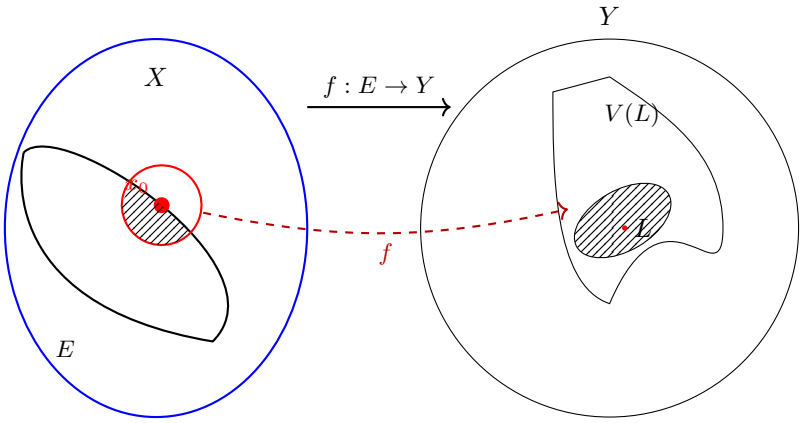


Figure 1: The definition of a limit. Left: the set E (black curve) inside X, the point x_0 (red), a neighborhood $U(x_0)$ (red circle) and the hatched set $U(x_0) \cap E$. Right: a neighborhood $V(L)$ of L in Y, which must contain the hatched image $f(U(x_0) \cap E)$.

DEFINITION 2.10. (Hausdorff) | A topological space Y is *Hausdorff* if any two distinct points $L \neq L'$ have disjoint neighborhoods $V(L) \cap V(L') = \emptyset$.

Lemma 2.11. (uniqueness of limits) | If Y is Hausdorff, a limit $\lim_{x \to x_0,\, x \in E} f(x)$ is unique when it exists.

Sequential limits. A sequence in Y is a function $x : \mathbb{N} \to Y$, $n \mapsto x_n$. By Lemma 2.8, ∞ is adherent to $E = \mathbb{N}$ inside $X = \mathbb{N} \cup \{\infty\}$, so Definition 2.9 applies with $x_0 = \infty$.

Corollary 2.12. (sequential limit) | For a sequence $x : \mathbb{N} \to Y$,

$$\lim_{\substack{n \to \infty \\ n \in \mathbb{N}}} x_n = L \iff \forall V(L),\ \exists N \in \mathbb{N} \mid \forall n > N,\ x_n \in V(L).$$

If the topology of Y is spanned by a basis, it is enough to let $V(L)$ range over basic sets containing L.

This is the only kind of limit we will need.

3 Metric Spaces

We need a more abstract notion of a metric space, since we do not have the real numbers yet: distances cannot be real numbers. Instead they take values in a set V in which we can add, subtract and compare. The examples we need are $V = \mathbb{Q}$ and, later, $V = \bar{\mathbb{Q}}$.

3.1 Well-valued sets

DEFINITION 3.1. (well-valued set) | A *well-valued set* is a set V with an operation $+$ and a relation $<$ such that

(V1) $(V,+)$ is an abelian group (we write 0 for the identity and $-a$ for inverses);
(V2) $<$ is a total order satisfying the compatibility implication

$$a < b \;\Rightarrow\; a + c < b + c \qquad \text{for all } a, b, c \in V;$$

(V3) V is *Archimedean*: for all $a, b > 0$ there exists an $n \in \mathbb{N}$ with $na > b$, where $na := a + \cdots + a$ (n times).

I furthermore use the following definitions:

- V is *dense* if for all $a < b$ in V there is a $c \in V$ with $a < c < b$;
- V is *nontrivial* if $V \neq \{0\}$;
- $|a| := \max(a, -a)$; $\quad a \leq b$ means $a < b$ or $a = b$.

Note: We don't actually need to assume communitivty since (V2) & (V3) imply it.$(a + b < (m + n + 1)\epsilon, b + a \geq (m + n)\epsilon \implies a + b - (b + a) < \epsilon$ for all epsilon greater than zero.

The Archimedean property (V3) is what allows us to build a sequence tending to zero (Lemma 3.16). Actually, for the completeness proofs we only need the *existence* of such a sequence; but with only that assumption we can end up constructing things that are not $\mathbb{R}$. This is still interesting, and we come back to it in Section 8.

Remark 3.2. (positivity) | Given (V1), condition (V2) is equivalent to a notion of *positivity*: a set $P \subseteq V$ (the positive elements) with

$$\text{(P1)}\;\; P + P \subseteq P, \qquad \text{(P2)}\;\; \forall a \in V \text{ exactly one of } a \in P,\; a = 0,\; -a \in P.$$

$$\begin{aligned}
\text{(V2)} \Rightarrow \text{(P)}: \quad & P := \{a \mid a > 0\}. \qquad a, b > 0 \;\Rightarrow\; a + b > 0 + b = b > 0 \;\; \text{(P1)}; \\
& a > 0 \iff 0 > -a \;\; (\text{add } -a), \text{ so (P2) is totality of } <. \\
\text{(P)} \Rightarrow \text{(V2)}: \quad & a < b \;:\iff b - a \in P. \qquad c - a = (c - b) + (b - a) \;\; \text{(transitive)}; \\
& \text{(P2)} \;\; \text{(total)}; \qquad (b + c) - (a + c) = b - a \;\; \text{(compatible)}.
\end{aligned}$$

This is how we will put an order on $\bar{\mathbb{Q}}$.

Lemma 3.3. (arithmetic of inequalities) | In a set satisfying (V1) and (V2):

(a) $a < b,\; c \leq d \;\Rightarrow\; a + c < b + d$;
(b) $a < b \iff -b < -a$;
(c) $|a| \geq 0;\;\; |a| = 0 \iff a = 0;\;\; |-a| = |a|;\;\; |a + b| \leq |a| + |b|;\;\; \big||a| - |b|\big| \leq |a - b|$;
(d) for $n \geq 1$: $na < nb \iff a < b$.

Lemma 3.4. (halving) | If V is dense: $\;\epsilon > 0 \;\Rightarrow\; \exists \eta > 0$ with $\eta + \eta \leq \epsilon$.

Notation. For $\epsilon > 0$ in a dense V we write $\epsilon/2$ for an element $\eta > 0$ as in Lemma 3.4, and $\epsilon/4$ for $(\epsilon/2)/2$, so $4(\epsilon/4) \leq \epsilon$; we write $\epsilon/3$ for $\epsilon/4$, which satisfies $3(\epsilon/3) < \epsilon$. When $V = \mathbb{Q}$ these are literally $\epsilon/2$, $\epsilon/4$ and (one may take) $\epsilon/3$. All we ever use is

$$\epsilon/2 + \epsilon/2 \leq \epsilon, \qquad 3(\epsilon/3) \leq \epsilon, \qquad 4(\epsilon/4) \leq \epsilon, \qquad 0 < \epsilon/2 < \epsilon, \qquad \epsilon - \epsilon/2 \geq \epsilon/2.$$

3.2 Metric spaces and the metric topology

DEFINITION 3.5. (metric space) | Let V be a nontrivial set satisfying (V1) and (V2). A *metric space* with values in V is a set X with a function $d : X \times X \to V$ such that for all $x, y, z \in X$:

1. $d(x, y) \geq 0$, and $d(x, y) = 0 \iff x = y$,
2. $d(x, y) = d(y, x)$,
3. $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality).

The *open ball* of radius $\epsilon \in V$, $\epsilon > 0$, is $B_\epsilon(x) = \{y \in X \mid d(x, y) < \epsilon\}$.

Lemma 3.6. (metric topology) | The open balls form a basis. The topology they span, the *metric topology*, is

$$\{\, U \subseteq X \mid \forall x \in U,\ \exists \epsilon > 0 \mid B_\epsilon(x) \subseteq U \,\},$$

which is the usual definition of an open set. Consequently, for sequences, (for the case of dense V, use triangle inequality to show, given an open ball containing L, we can make a open ball centered at L contained in the former ball. For the non-dense V, the sequence must be eventaully constant so the definitions coincide.)

$$\lim_{n\to\infty} x_n = L \iff \forall \epsilon \in V,\ \epsilon > 0,\ \exists N \mid \forall n > N,\ d(x_n, L) < \epsilon.$$

We write d- lim when we want to name the metric being used.

Lemma 3.7. (metric spaces are Hausdorff) | Let V Every metric space with values in V is Hausdorff. Hence limits in such spaces are unique (Lemma 2.11). (treat each case separately: V is dense, V is not dense & non-trivial, V is trivial)

DEFINITION 3.8. (natural metric of a well-valued set) | Any well-valued set V inherits the following natural metric, with values in V itself:

$$d : V \times V \to V, \qquad d(x, y) = |x - y|.$$

Lemma 3.9. $d(x, y) = |x-y|$ is a metric on V, and its metric topology is the order topology of Theorem 2.7.

3.3 Cauchy sequences and holes

DEFINITION 3.10. (Cauchy sequence, completeness) | A sequence (x_n) in a metric space (X, d) is *Cauchy* if

$$\forall \epsilon > 0,\ \exists N \mid \forall n > N,\ d(x_n, x_N) < \epsilon.$$

(X, d) is *complete* if every Cauchy sequence in X converges to a point of X.

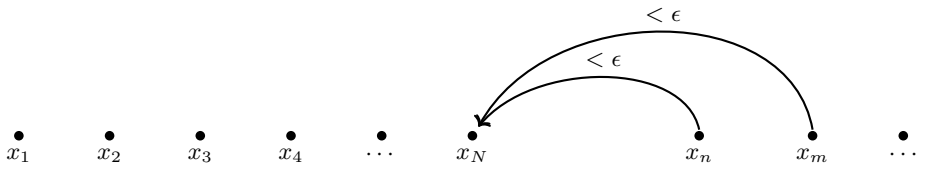


Figure 2: The Cauchy condition: all sufficiently late terms lie within ϵ of the fixed term x_N.

This is how the condition is used in proofs: to compare two late terms x_n, x_m we move each of them to x_N, $d(x_n, x_m) \leq d(x_n, x_N) + d(x_N, x_m) < \epsilon + \epsilon$, so we only ever measure distances to this one x_N (Figure **??**).

Lemma 3.11. (convergent $\Rightarrow$ Cauchy) | Let V Every convergent sequence is Cauchy.

Lemma 3.12. (Cauchy sequences are bounded) | (x_n) Cauchy $\Rightarrow$ $\exists a \in X,\ M \in V:\ \forall n,\ d(x_n, a) \leq M$.

A *hole* is the failure of the converse of Lemma 3.11: a Cauchy sequence that tries to converge but has nothing in the space to converge to. Every Cauchy sequence is bounded (Lemma 3.12), and much more is true: for each ϵ its whole tail beyond some N is squeezed inside the ball $B_\epsilon(x_N) = (x_N - \epsilon,\ x_N + \epsilon)$, and we may make ϵ as small as we like. So the tail is trapped in smaller and smaller balls, and any Cauchy sequence seems to "converge" to something. When it does not, the place it is squeezing down on is missing from the space, and we call it a *hole* (Figure 3).

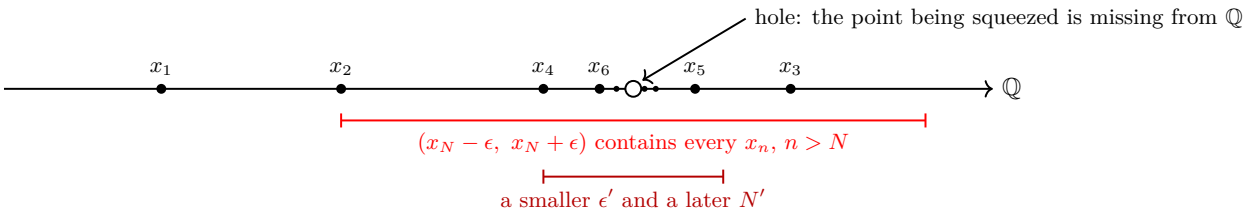


Figure 3: A hole. The terms of a Cauchy sequence get trapped in smaller and smaller intervals (red). They squeeze down on a single location, but if that location (open circle) is not a point of the space, the sequence has nothing to converge to.

There exist well-valued sets with holes!

THEOREM 3.13. (Q has holes) | The rational numbers with their natural metric $d(x, y) = |x - y|$ are not complete.

Lemma 3.14. (non-dense well-valued sets have no holes) | This is trivially true in the trivial case since the ambient space can only have one element. So, let V be a nontrivial well-valued set that is *not* dense. Then V has a smallest positive element p, $V = \{kp \mid k \in \mathbb{Z}\}$ is a copy of the integers, and every metric space with values in V is complete.

Henceforth we are only concerned with dense well-valued sets; by Lemma 3.14 nothing is lost. Such sets I like to call $\mathbb{Q}$-*like*. The goal is to construct a completion of these sets.

(In looking for simular uses of such well-valued sets, I note Hölders theorem which says any such well valued set is a subgroup of the reals. Such a result is a simple implication of the results of the completeness proof of this paper with lemma 3.14 In essense we have already considered the simple cases of when V is not dense or trivial... now we consider the interesting case...)

DEFINITION 3.15. ($\mathbb{Q}$-like set) | A $\mathbb{Q}$-*like set* is a dense, nontrivial well-valued set.

Lemma 3.16. (a sequence tending to zero) | Let V be $\mathbb{Q}$-like. There is a sequence (e_n) in V with

$$e_n > 0, \qquad e_{n+1} + e_{n+1} \leq e_n, \qquad \forall \epsilon > 0\ \exists N\ \forall n \geq N,\ e_n < \epsilon.$$

(such a sequence looks like $e_n = (\frac{1}{2})^n$.)

4 The Problem

Let $\mathbb{Q}$ be a $\mathbb{Q}$-like set (i.e. dense,ordered abelian group with archemedian property that contains more than just the zero element –Definition 3.15); We consider $\mathbb{Q}$ with its natural metric

$$d : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q} \mid d(x, y) = |x - y|,$$

whose topology is the order topology (Lemma 3.9). Our problem is that this metric need not be complete: for the rationals it has holes (Theorem 3.13). We therefore wish to find a larger metric space $\bar{\mathbb{Q}}$ such that

(1) $(\mathbb{Q}, d) \subseteq (\bar{\mathbb{Q}}, \bar{d})$,

(2) $(\bar{\mathbb{Q}}, \bar{d})$ is complete,

(3) if Z is any complete metric space with $(\mathbb{Q}, d) \subseteq (Z, d_Z)$, then $(\bar{\mathbb{Q}}, \bar{d}) \subseteq (Z, d_Z)$.

These properties make $\bar{\mathbb{Q}}$ the set containing $\mathbb{Q}$ (1), all of the missing holes (2), and nothing else (3). Here "$\subseteq$" means "embeds isometrically into"; this is made precise in Section 4.2 and Section 7. I now construct such a set and prove it has the desired properties.

4.1 Formal Construction of $\bar{\mathbb{Q}}$

The plan: construct $\bar{\mathbb{Q}}$ as equivalence classes of Cauchy sequences (Step 1); construct $\bar{d} : \bar{\mathbb{Q}} \times \bar{\mathbb{Q}} \to \bar{\mathbb{Q}}$ and show it is well defined (Step 2); show that $\bar{\mathbb{Q}}$ is itself a valid set of values, i.e. a $\mathbb{Q}$-like set (Step 3); and show that $\bar{d}$ is a metric, in fact the natural one (Step 4).

Step 1 | Construct the ambient space.

Write $\mathscr{L}(\mathbb{N} \to \mathbb{Q})$ for the set of all functions $\mathbb{N} \to \mathbb{Q}$, i.e. all sequences in $\mathbb{Q}$. It exists by the power set axiom: a function is a subset of $\mathbb{N} \times \mathbb{Q}$, so $\mathscr{L}(\mathbb{N} \to \mathbb{Q})$ is carved out of $\mathcal{P}(\mathbb{N} \times \mathbb{Q})$ by specification. By specification again we keep only the Cauchy sequences:

$$S := \{(x_n) \in \mathscr{L}(\mathbb{N} \to \mathbb{Q}) \mid (x_n) \text{ is a Cauchy sequence in } (\mathbb{Q}, d)\}.$$

Define the relation on S:

$$(x_n) \sim (y_n) \iff d\text{-}\lim_{\substack{n \to \infty \\ n \in \mathbb{N}}} d(x_n, y_n) = 0.$$

Lemma 4.1. $\sim$ is an equivalence relation on S.

This defines equivalence classes:

$$[(x_n)] = \{(z_n) \in S \mid (z_n) \sim (x_n)\}, \qquad [(x_n)] = [(y_n)] \iff (x_n) \sim (y_n).$$

By the axiom of replacement (or by specification inside $\mathcal{P}(S)$), we replace each $(x_n) \in S$ with its equivalence class $[(x_n)]$. Thus the ambient space is:

$$\bar{\mathbb{Q}} := \{[(x_n)] \mid (x_n) \in \mathscr{L}(\mathbb{N} \to \mathbb{Q}),\ (x_n) \text{ Cauchy}\}.$$

Step 2 | Define the distance.

$$\bar{d} : \bar{\mathbb{Q}} \times \bar{\mathbb{Q}} \to \bar{\mathbb{Q}}, \qquad \bar{d}([(x_n)], [(y_n)]) := [(d(x_n, y_n))].$$

I must show that (a) $[(d(x_n, y_n))]$ is indeed in $\bar{\mathbb{Q}}$ and (b) that this is a well defined function ...

Step 3 | $\bar{\mathbb{Q}}$ is a $\mathbb{Q}$-like set.

Before we can even say that $\bar{d}$ is a metric we need "+" and "<" on $\bar{\mathbb{Q}}$, satisfying (V1) and (V2). We also check (V3), density and nontriviality, so that $\bar{\mathbb{Q}}$ is as good a set of values as $\mathbb{Q}$ was.

a | (V1) abelian group. Define

$$[(x_n)] + [(y_n)] := [(x_n + y_n)], \qquad -[(x_n)] := [(-x_n)], \qquad 0 := [(0, 0, 0, \dots)].$$

The abelian group laws hold because they hold term by term in $\mathbb{Q}$. For $a \in \mathbb{Q}$ we write $\iota(a) := [(a, a, a, \dots)]$ for the class of the constant sequence; I come back to this in section 4.2. Note $\iota(a) + \iota(b) = \iota(a + b)$ and $-\iota(a) = \iota(-a)$.

b | (V2) positivity and order.

DEFINITION 4.2. (positivity, order) | Call $[(x_n)]$ *positive*, written $[(x_n)] > 0$, if

$$\exists \delta \in \mathbb{Q},\ \delta > 0,\ \exists N \mid \forall n > N,\ x_n \geq \delta.$$

Define $X < Y \iff Y - X > 0$, and $X \leq Y \iff X < Y$ or $X = Y$.

Lemma 4.3. (trichotomy) | For every $X = [(x_n)] \in \bar{\mathbb{Q}}$ exactly one of $X = 0$, $X > 0$, $-X > 0$ holds. Consequently $\bar{\mathbb{Q}}$ satisfies (V1) and (V2).

The following lemma is the bridge between orderings of sequences and orderings of classes; we use it constantly.

Lemma 4.4. (eventual comparison) | For $(u_n), (v_n) \in S$:

$$\exists N \mid \forall n > N, u_n \leq v_n \ \Rightarrow\ [(u_n)] \leq [(v_n)].$$

Lemma 4.5. ($\mathbb{Q}$ is coinitial and cofinal in $\bar{\mathbb{Q}}$) |
(a) $\epsilon \in \bar{\mathbb{Q}},\ \epsilon > 0 \ \Rightarrow\ \exists q \in \mathbb{Q} :\ 0 < \iota(q) < \epsilon$.
(b) $X \in \bar{\mathbb{Q}} \ \Rightarrow\ \exists q \in \mathbb{Q} :\ X < \iota(q)$.

c | (V3) Archimedean.

$$X, Y > 0 \ \Rightarrow\ x_n \geq \delta > 0\ (n \gg 0) \ \Rightarrow\ X \geq \iota(\delta); \qquad Y < \iota(M) \ \text{ for some } M \in \mathbb{Q} \ \ (4.5\text{b});$$
$$\text{(V3) in } \mathbb{Q} \ \Rightarrow\ \exists n :\ n\delta > M \ \Rightarrow\ nX \geq n\,\iota(\delta) = \iota(n\delta) > \iota(M) > Y.$$

d | density, nontriviality.

$$X < Y \ \Rightarrow\ Y - X > 0 \xRightarrow{4.5\text{a}} \exists q \in \mathbb{Q} :\ 0 < \iota(q) < Y - X \ \Rightarrow\ X < X + \iota(q) < Y.$$

$$c > 0 \text{ in } \mathbb{Q} \ \Rightarrow\ \iota(c) > 0 \ \Rightarrow\ \bar{\mathbb{Q}} \neq \{0\}.$$

Step 4 | $(\bar{\mathbb{Q}}, \bar{d})$ is a metric space with values in $\bar{\mathbb{Q}}$, and $\bar{d}$ is the natural metric of $\bar{\mathbb{Q}}$.

Lemma 4.6. ($\bar{d}$ is the natural metric of $\bar{\mathbb{Q}}$) | For all $X, Y \in \bar{\mathbb{Q}}$: $\bar{d}(X, Y) = |Y - X|$, with $|\cdot|$ taken in the $\mathbb{Q}$-like set $\bar{\mathbb{Q}}$. So $\bar{\mathbb{Q}}$ carries exactly the same kind of metric as $\mathbb{Q}$ did (Definition 3.8), and its topology is the order topology (Lemma 3.9).

4.2 Isometric embedding

Map each $a \in \mathbb{Q}$ to the class of the constant sequence $(a, a, a, \dots)$:

$$\iota : \mathbb{Q} \to \bar{\mathbb{Q}}, \qquad \iota(a) := [(a, a, a, \dots)].$$

$$\iota(a) = \iota(b) \;\Rightarrow\; d\text{-}\lim_{n\to\infty} |a - b| = 0 \;\Rightarrow\; |a - b| = 0 \;\Rightarrow\; a = b \qquad \text{(injective)}$$

$$\iota(a) + \iota(b) = \iota(a + b); \qquad a < b \iff \iota(a) < \iota(b) \qquad \text{(Step 3a, 3b)}$$

$$\bar{d}(\iota(a), \iota(b)) = [(d(a, b), d(a, b), \dots)] = \iota(d(a, b)) \qquad \text{(isometric)}$$

Hence ι is an isometric embedding $(\mathbb{Q}, d) \subseteq (\bar{\mathbb{Q}}, \bar{d})$, and property (1) holds. From now on we freely regard $\mathbb{Q}$ as a subset of $\bar{\mathbb{Q}}$, writing a for $\iota(a)$ when no confusion is possible.

Distances in $\bar{\mathbb{Q}}$ are elements of $\bar{\mathbb{Q}}$, so an "ϵ" in $\bar{\mathbb{Q}}$ is a class, not a number of $\mathbb{Q}$. The next corollary says that this never matters: it is enough to use ϵ's coming from $\mathbb{Q}$.

Corollary 4.7. (wiseness) | For $Y_n, Y \in \bar{\mathbb{Q}}$,

$$\forall \epsilon \in \mathbb{Q},\ \epsilon > 0 :\quad \bar{d}(Y_n, Y) < \iota(\epsilon)\ (n \gg 0) \qquad \iff \qquad \forall \epsilon \in \bar{\mathbb{Q}},\ \epsilon > 0 :\quad \bar{d}(Y_n, Y) < \epsilon\ (n \gg 0).$$

We call the left side $\mathbb{Q}$-*wise* and the right side $\bar{\mathbb{Q}}$-*wise* convergence; the same holds for the Cauchy condition. In particular $\iota(e_n) \to 0$ in $\bar{\mathbb{Q}}$.

5 Completeness proof via Sequences

THEOREM 5.1. (completeness) | $(\bar{\mathbb{Q}}, \bar{d})$ is complete.

The idea of this proof is that the real numbers are themselves sequences. The core idea is the following:

$$\boxed{\bar{d}\big([a_n],[b_n]\big) < l(\epsilon_q) \;\Rightarrow\; \exists N \mid \forall n \geq N,\ |a_n - b_n| < \epsilon_q}$$

$$\begin{array}{lccccccccc} [a_n]\colon & a_1 & a_2 & a_3 & \cdots & a_N & a_{N+1} & a_{N+2} & a_{N+3} & \cdots \\ & & & & & \big| < \epsilon_q & \big| < \epsilon_q & \big| < \epsilon_q & \big| < \epsilon_q & \\ [b_n]\colon & b_1 & b_2 & b_3 & \cdots & b_N & b_{N+1} & b_{N+2} & b_{N+3} & \cdots \end{array}$$

$$\underbrace{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad}_{\text{tail}}$$

Let (X_n) be cauchy in $(\bar{\mathbb{Q}}, \bar{d})$. Use choice to get representatives (a_n) of each X_n (i.e. each $X_n = [(a_n)]$). Explicit constructions of these representatives are, however, possible**

5.1 Construct two sequences

Step 1 | $(N_k) : \mathbb{N} \to \mathbb{N}$.

(i) $\forall n > N_k, \quad \bar{d}\Big(\big[a^{(N_k)}\big],\ \big[a^{(n)}\big]\Big) < e_k$;

(ii) N_k is the minimal such N.

2nd idea for sequences. This implies that the tails are eventually all e_k-close. This is exactly how we choose our j_k...

Step 2 | $(j_k) : \mathbb{N} \to \mathbb{N}$.

(i) $j_1 = 1$ *(doesnt rlly matter)*

(ii) $\forall j \geq j_k, \quad d\big(a_{j_k}^{(N_k)},\ a_j^{(N_k)}\big) < e_k$ *(cauchy elements)*

(iii) $\forall j \geq j_{k+1}, \quad d\Big(a_{j_{k+1}}^{(N_{k+1})},\ a_{j_{k+1}}^{(N_k)}\Big) < e_{k+1}$; *(core idea)*

(iv) $j_{k+1} \geq j_k$; *(monotomically increasing)*

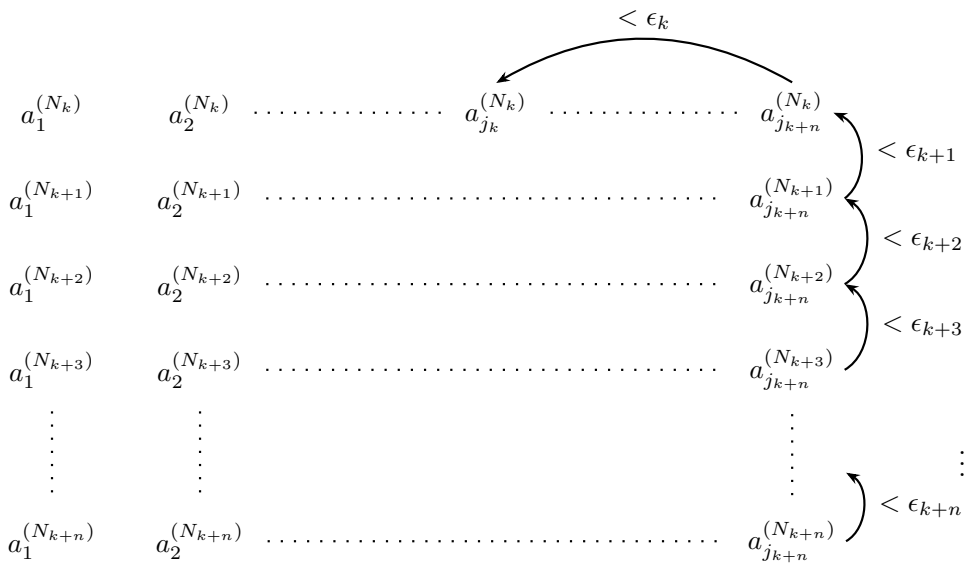
(v) j_{k+1} is minimal subject to (ii) & (iii) & (iv) *(no choice).*

As I now show, it intuitively follows from a picture of our cauchy sequence as a sequence of sequences that $\big[(a_{j_k}^{(N_k)})_{k=1}^{\infty}\big]$ gives us an element of $\bar{\mathbb{Q}}$ that the sequence (X_n) converges to in the $(\bar{\mathbb{Q}}, \bar{d})$ metric.

Lemma 5.2. (geometric series) | For all $B \geq 0$ and $m \geq 1$, $e_B + e_{B+1} + \cdots + e_{B+m-1} \;\leq\; 2e_B - e_{B+m-1} \;<\; 2e_B$.

Step 1 | $[(a_{j_k}^{(N_k)})_{k\in\mathbb{N}}]$ is cauchy ...

We prove this by method of drawing a hangman



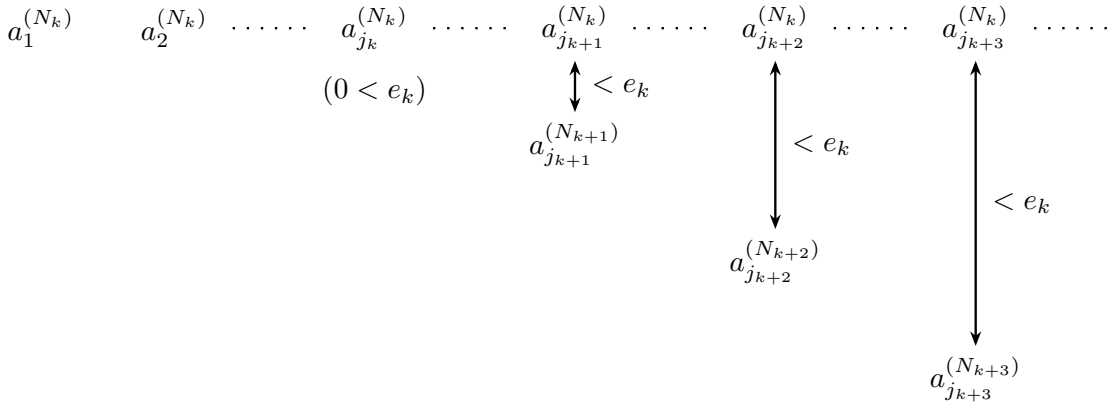
$$d\big(a_{j_k}^{(N_k)},\, a_{j_{k+n}}^{(N_{k+n})}\big) < \sum_{m=0}^{n} e_{k+m}$$

which, by our geometric series lemma, will be, for ALL n, ...

$$< 2e_k$$

Step 2 | prove convergence

It suffices to show that $[(a^{(N_k)}]$ is $l(2e_k)$ close to $\big[(a_{j_k}^{(N_k)})_{k\in\mathbb{N}}\big]$. Because $[(a^{(N_k)}]$ is cauchy, it will suffice to show that a subsequence of it is eventually (e_k) close. Simply consider $(a_{j_{k+n}}^{N_k})_{n\in\mathbb{N}}$ and the result immediately follows from our core idea...



Note that we actually don't even need step one because a sequence equivalent to a cauchy one is automatically cauchy. All we need is the proof the $< e_k$ terms which is already apparent from the same diagram of a sequence of sequences. The entire proof is contained in this diagram.

6 Completeness proof via points

I prove the same complete

THEOREM 6.1. (formal limits are real limits) | Let (a_n) be a Cauchy sequence in $(\mathbb{Q}, d)$. Then, in $(\bar{\mathbb{Q}}, \bar{d})$,

$$\bar{d}\text{-}\lim_{\substack{n\to\infty \\ n\in\mathbb{N}}} \iota(a_n) = [(a_n)].$$

I.e. once viewed inside $\bar{\mathbb{Q}}$, a Cauchy sequence converges, and its limit is its own class. It is this theorem that beautifully allows us to turn Cauchy sequences into limits.

Proof. By Corollary 4.7 it suffices to take $\epsilon \in \mathbb{Q}$, $\epsilon > 0$. Choose N so large that $d(a_n, a_N) < \epsilon/2$ for all $n > N$. Then, for all $m > N$,

$$\begin{aligned}
\bar{d}\big(\iota(a_m), [(a_n)]\big) &\le \bar{d}\big([a_m\ a_m\ \dots], [a_N\ a_N\ \dots]\big) + \bar{d}\big([a_N\ a_N \dots], [(a_n)]\big) && \text{(triangle inequality in } \bar{\mathbb{Q}}\text{, Step 4)} \\
&= \iota\big(d(a_m, a_N)\big) + \big[(d(a_N, a_n))_{n\in\mathbb{N}}\big] && (\iota \text{ is an isometry; definition of } \bar{d}) \\
&< \iota(\epsilon/2) + \iota(\epsilon/2) && \text{(3b; Lemma 4.4)} \\
&\le \iota(\epsilon). && \square
\end{aligned}$$

THEOREM 6.2. ($\mathbb{Q}$ is dense in $\bar{\mathbb{Q}}$) |

$$\forall \epsilon \in \bar{\mathbb{Q}},\ \epsilon > 0, \quad \forall X \in \bar{\mathbb{Q}}, \quad \exists q \in \mathbb{Q} \mid \bar{d}(\iota(q), X) < \epsilon.$$

Proof. Choose a representative (a_n) of X, i.e. $X = [(a_n)]$; this is the only non-explicit choice in the proof. By Lemma 4.5(a) there is an $\epsilon_q \in \mathbb{Q}$ with $0 < \iota(\epsilon_q) < \epsilon$. Choose N (explicitly, say the least one) such that $d(a_n, a_N) < \epsilon_q$ for all $n > N$, and put $q := a_N$. Then

$$\begin{aligned}
\bar{d}\big(\iota(q), X\big) &= \big[(d(a_N, a_n))_{n\in\mathbb{N}}\big] && \text{(definition of } \bar{d}) \\
&\le \big[(\epsilon_q, \epsilon_q, \epsilon_q, \dots)\big] && \text{(Lemma 4.4, as } d(a_N, a_n) < \epsilon_q \text{ for } n > N) \\
&= \iota(\epsilon_q) \\
&< \epsilon. && \square
\end{aligned}$$

THEOREM 6.3. (completeness) | $(\bar{\mathbb{Q}}, \bar{d})$ is complete.

We wish to prove completeness via the same as 6.1, but the issue is each Cauchy sequence may not be in $\mathbb{Q}$. The idea (Figure 4): replace each term of the given Cauchy sequence by a nearby element of $\mathbb{Q}$. These form a Cauchy sequence *in* $\mathbb{Q}$, so their class is a point of $\bar{\mathbb{Q}}$, and by Theorem 6.1 it is their limit, hence also the limit of the original sequence.

Proof. Let (X_n) be a Cauchy sequence in $(\bar{\mathbb{Q}}, \bar{d})$, and let (e_n) be the null sequence of Lemma 3.16; by Corollary 4.7, $\iota(e_n) \to 0$ in $\bar{\mathbb{Q}}$. By Theorem 6.2, for each n choose

$$q_n \in \mathbb{Q} \quad \text{with} \quad \bar{d}\big(X_n, \iota(q_n)\big) < \iota(e_n).$$

Then (q_n) is a Cauchy sequence in $\mathbb{Q}$ (Part I), so $L := [(q_n)]$ is a point of $\bar{\mathbb{Q}}$, and $X_n \to L$ (Part II).

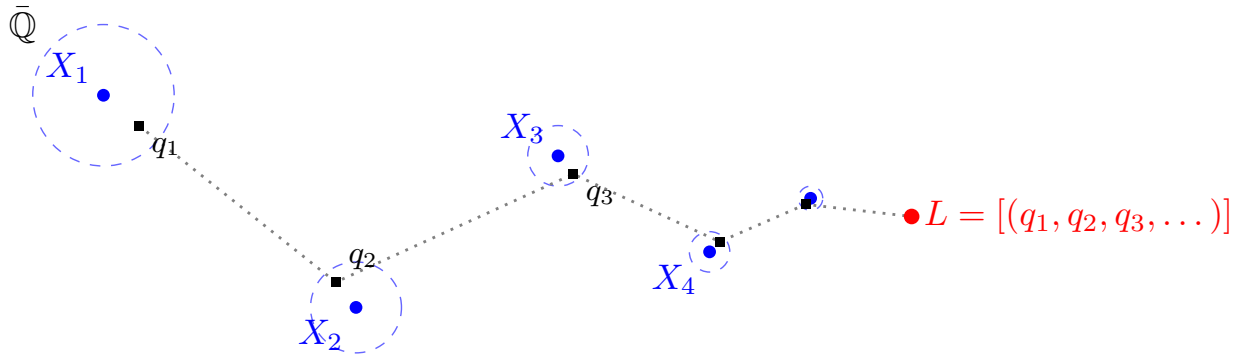


Figure 4: The idea of the proof. Blue: the given Cauchy sequence (X_n) in $\bar{\mathbb{Q}}$, each term with a dashed ball of radius $\iota(e_n)$. Black squares: an element $q_n \in \mathbb{Q}$ chosen inside each ball (Theorem 6.2). The q_n form a Cauchy sequence *in* $\mathbb{Q}$, so they define a point $L = [(q_n)]$ of $\bar{\mathbb{Q}}$, and by Theorem 6.1 the q_n converge to L. Since X_n stays within $\iota(e_n)$ of q_n, the X_n converge to L too.

Part I | (q_n) is a Cauchy sequence in $(\mathbb{Q}, d)$. Let $\epsilon \in \mathbb{Q}$, $\epsilon > 0$, and put $\epsilon' := \iota(\epsilon)/3 \in \bar{\mathbb{Q}}$. Choose N so large that

$$\begin{aligned} &(1) \quad \bar{d}(X_n, X_N) < \epsilon' \ \text{ for all } n > N && ((X_n) \text{ is Cauchy}),\\ &(2) \quad \iota(e_n) < \epsilon' \ \text{ for all } n \geq N && (\iota(e_n) \to 0). \end{aligned}$$

Then, for all $n > N$,

$$\begin{aligned} \iota\big(d(q_n, q_N)\big) &= \bar{d}\big(\iota(q_n), \iota(q_N)\big) && (\iota \text{ is an isometry})\\ &\leq \bar{d}\big(\iota(q_n), X_n\big) + \bar{d}\big(X_n, X_N\big) + \bar{d}\big(X_N, \iota(q_N)\big) && (\text{triangle inequality in } \bar{\mathbb{Q}})\\ &< \iota(e_n) + \epsilon' + \iota(e_N) && (\text{choice of } q_n,\ q_N;\ (1))\\ &< \epsilon' + \epsilon' + \epsilon' && ((2))\\ &\leq \iota(\epsilon), \end{aligned}$$

and therefore $d(q_n, q_N) < \epsilon$, because ι reflects order (Step 3b).

Part II | $X_n \to L$. Let $\epsilon \in \bar{\mathbb{Q}}$, $\epsilon > 0$. Choose N so large that

$$\begin{aligned} &(1) \quad \bar{d}(X_n, X_N) < \epsilon/3 \ \text{ for all } n > N && ((X_n) \text{ is Cauchy}),\\ &(2) \quad \iota(e_N) < \epsilon/3 && (\iota(e_n) \to 0),\\ &(3) \quad \bar{d}\big(\iota(q_N), L\big) < \epsilon/3 && (\text{Theorem 6.1: } \iota(q_n) \to L). \end{aligned}$$

The proof then comes down to a single line usage of the triangle inequality.....
for all $n > N$,

$$\begin{aligned} \bar{d}\big(X_n, L\big) &\leq \bar{d}\big(X_n, X_N\big) + \bar{d}\big(X_N, \iota(q_N)\big) + \bar{d}\big(\iota(q_N), L\big) && (\text{triangle inequality in } \bar{\mathbb{Q}})\\ &< \epsilon/3 + \iota(e_N) + \epsilon/3 && ((1);\ \text{choice of } q_N;\ (3))\\ &< \epsilon/3 + \epsilon/3 + \epsilon/3 && ((2))\\ &\leq \epsilon. && \square \end{aligned}$$

7 Smallest complete metric

To make property (3) precise we must say what a "complete metric space containing $\mathbb{Q}$" is. The idea is that we have some larger metric space (Z, d_Z) with values in some set A that contains Q. Then, this metric space "contains" Q if it contains (Q, d) isomorphically. This makes sense because the values A contain Q and therefore we can have values from d_Z isomorphically equivalent to values from d. More precisely, we make the following thereom....

DEFINITION 7.1. (isometric imbedding) | Let (Q, d) be a metric space with values in abelian and compatibly ordered C. Let (Z, d_Z) be a complete metric space with values in abelian and compatibly ordered A. We say that $(Q, d) \subseteq (Z, d_Z)$ if (1) there exists an isomorphic mapping l taking $(C, +, <)$ into $(A, +, <)$, (2) there exists an isomorphic mapping ϕ taking $\mathbb{Q}$ into Z, and (3) the distance $d_Z : Z \times Z \to A$ is such that $\forall q_1, q_2 \in Q \; l(d(q_1, q_2)) = d_Z(\phi(q_1), \phi(q_2))$.

THEOREM 7.2. (smallest complete metric) | Let (Z, d_Z) be a complete metric space with values in $(A, +, <)$. If $(Q, d) \subseteq (Z, d_Z)$, then $(\bar{\mathbb{Q}}, \bar{d}) \subseteq (Z, d_Z)$. The isomorphic imbedding of values into A is assumed implicitly by the usual manner in accordance with the definition of an isometric imbedding.

8 Using only a sequence to zero

Looking back, the Archimedean property (V3) of $\mathbb{Q}$ was used in only two places: to produce the null sequence (e_n) (Lemma 3.16), and to prove (V3) for $\bar{\mathbb{Q}}$ (Step 3c). Everything else used only the existence of (e_n). So (V3) may be replaced by the weaker assumption

(V3′) there is a sequence $e_n > 0$ in V with $\forall \epsilon > 0\ \exists N\ \forall n \geq N,\ e_n < \epsilon$;

such a V is automatically dense (there is no smallest positive element), and (e_n) may be chosen with $e_{n+1} + e_{n+1} \leq e_n$ (replace e_{n+1} by $\min(e_{n+1}, e_n/2)$).

THEOREM 8.1. (completion without Archimedes) | Let $\mathbb{Q}$ be nontrivial and satisfy (V1), (V2) and (V3′). Then Sections 4.1–7 hold verbatim, except that $\bar{\mathbb{Q}}$ satisfies (V3′), with null sequence $\iota(e_n)$ (Corollary 4.7), in place of (V3).

The completion need no longer be $\mathbb{R}$:

Example 8.2. ($\mathbb{Q} \times \mathbb{Q}$ completes to $\mathbb{Q} \times \mathbb{R}$) | Let $V = \mathbb{Q} \times \mathbb{Q}$ (now $\mathbb{Q}$ the rationals) with componentwise addition and the *lexicographic* order: $(a, b) > 0$ iff $a > 0$, or $a = 0$ and $b > 0$. It satisfies (V1), (V2) and (V3′) with $e_n = (0, (\frac{1}{2})^n)$, but not (V3): $n \cdot (0, 1) = (0, n) < (1, 0)$ for every n. For a Cauchy sequence (a_n, b_n), taking $\epsilon = (0, 1)$ shows that a_n is eventually constant, say $= a$, and taking $\epsilon = (0, \delta)$ shows that (b_n) is Cauchy in $\mathbb{Q}$. Hence $[(a_n, b_n)] \mapsto (a, [(b_n)])$ identifies $\bar{\mathbb{Q}}$ with $\mathbb{Q} \times \mathbb{R}$ (lexicographic), which is not Archimedean.

9 Completing any metric space

Say I have some arbitrary metric $(X, d : X \times X \to A)$ for some well-valued A. We wish to find the smallest metric that completes this one. Under the strength of the contents of this paper hitherto, the completion of A is either isomorphic to $\mathbb{Z}$ (th integers) or to $\mathbb{R}$ (the reals, $\mathbb{Q}$). However the integers isomorphically map into the reals so there is no harm in just always choosing the resulting completion metric to take values in $\mathbb{R}$....
Our goal is, given a metric $(X, d_X : X \times X \to \mathbb{R})$, find a larger space $(\bar{X}, d_{\bar{X}} : \bar{X} \times \bar{X} \to \mathbb{R})$ that (1) is complete (2) contains isometrically (X, d), and (3) is the smallest set that does (1) & (2).

The result is immediate as the proofs are exactly the same. We define

$$\bar{X} = \{[(a_n)] \mid \quad (a_n) \text{ is cauchy sequence in } (X, d_X)\}$$
$$d_{\bar{X}}([(a_n)], [(b_n)]) = [(d_X(a_n, b_n))_{n \in \mathbb{N}}] = \lim_{n\ \in \mathbb{N}, n \to \infty} d_X(a_n, b_n)$$

The equivalence relation required in eq ___ is also the same as is required by our distance function needing to be a metric. Because nothing in our proofs of completeness used any of the structure on the metric space $\mathbb{Q}$ (only used this stuff on the values space), the same proofs follow through.

Note the reason for denoting $\bar{X}$ and $\bar{Q}$ is because these sets are the closure of the corresponding X and Q in their respective metric spaces.

10 What sets complete to R?

add multiplication to R. Everything follows through from added multiplication from Q and we define

$$XY := ...$$

The isormphism l also preserves this multiplication so it all works out. Now we have a notion of the number 1. Such a set $\bar{Q}, +, <, *, \bar{d}(X,Y) := |X - Y|$ is ussually denoted by the familiar $\mathbb{R}$.

example sets:

$$A := \{\frac{z}{2^n} \mid z \in \mathbb{Z}, n \in \mathbb{N}\}$$

Given $X \in \mathbb{R}, X \geq 0$ By archemedian, given an n,

$$\exists N \mid 2^n X < N$$

Explicitely choose the minimal such N and denote $N_{0,(n)} := N_{\text{minimal}} - 1$. We have,

$$N_{0,(n)} := \lfloor 2^n X \rfloor \leq 2^n X < N_{0,(n)} + 1$$

We get the resulting sequence $\frac{\lfloor 2^n X \rfloor}{2^n}$ which is a cauchy sequence into A and whose limit / generating equivalence class gives exactly X. This gives an explicit construction of a element of a real number without using choice, as promised.

decimal representation, base anything representation, etc. all just different versions of finding explicit formulas for the equivalence class of Cauchy sequences that makes a real number a real number.